

Week 8: Simulated Attack & Cloud Incident Response

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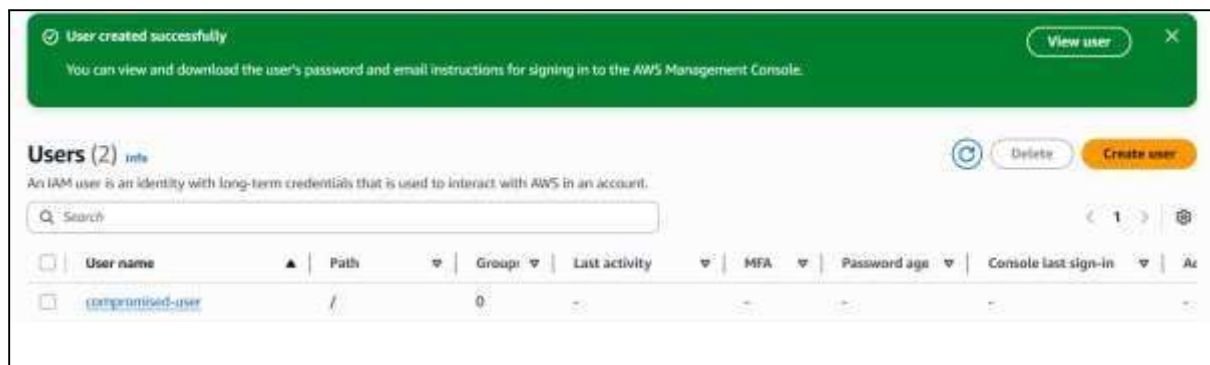
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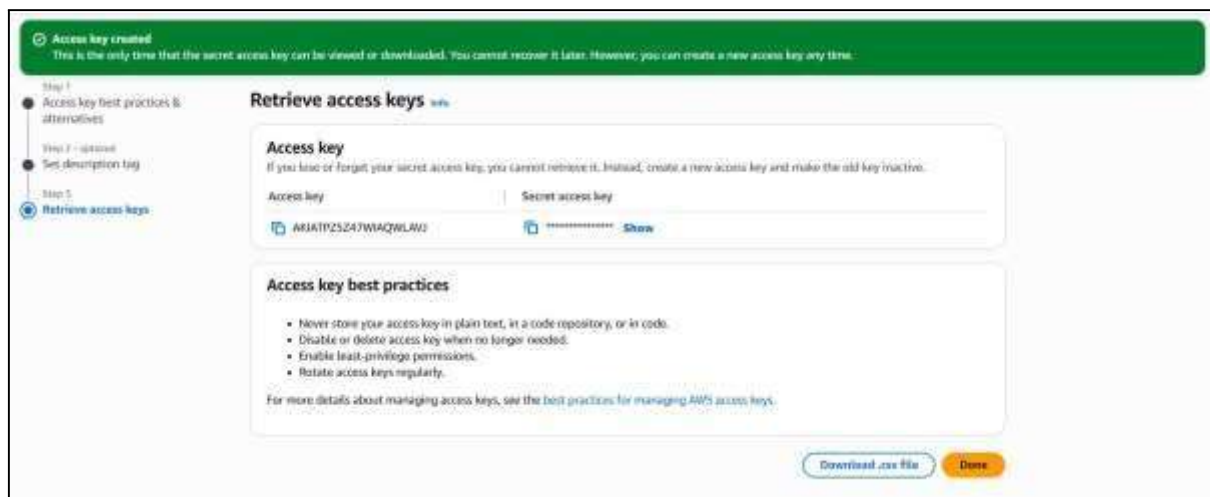
- Task-** 1) Simulate a compromised account scenario.
2) Respond & secure the account.
3) Document the incident response steps.

Task 1- Simulate a Compromised Account.

- 1) Log in to AWS Console. Go to <https://aws.amazon.com>, sign in to the AWS Management Console.
- 2) Create a test IAM user named **compromised-user** with basic permissions (e.g., Amazon S3 read access).



- 3) Generate **Access Keys** for the user.



- 4) Share these keys intentionally with a test machine/script that attempts unauthorized AWS CLI actions (e.g., trying to delete S3 objects without permission).

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[illegible]

5) Monitor AWS CloudTrail and CloudWatch to detect unusual activity from the user account.



Task 2- Respond to the Incident.

1) Identify the Source: Go to CloudTrail → Event History. Filter by compromised-user and check the activity timeline.

```
CloudShell
ap-south-1 +
~ $ aws cloudtrail lookup-events \
> --lookup-attributes AttributeKey=Username,AttributeValue=compromised-user \
~ $ aws cloudtrail lookup-events \
> --lookup-attributes AttributeKey=Username,AttributeValue=compromised-user \
> --max-results 50
{
  "Events": [
    {
      "EventId": "6b3b401f-7f0a-4fc6-9584-890cd06cbf8e",
      "EventName": "ListBuckets",
      "ReadOnly": "true",
      "AccessKeyId": "AKIAIPZ5Z47WIAQMLAVJ",
      "EventTime": "2025-08-18T08:19:54+00:00",
      "EventSource": "s3.amazonaws.com",
      "Username": "compromised-user",
      "Resources": [],
      "CloudTrailEvent": "{\"eventVersion\":\"1.11\",\"userIdentity\":{\"type\":\"IAMUser\"},\"principalId\":\"AIDAIPT25Z47WQ4PDXBV7\",\"arn\":\"arn:aws:iam:240110921708:user/compromised-user\", \"countId\":\"24:...skipping...\"}"
    }
  ]
}
```

2) Secure the Account: Deactivate the compromised Access Keys: `aws iam update-access-key --access-key-id <ACCESS_KEY> --status Inactive --user-name compromised-user`

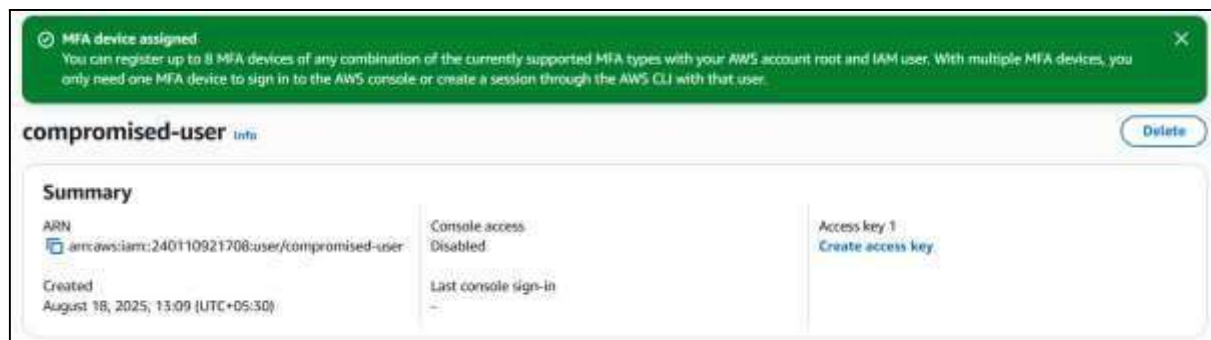
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```
~ $  
~ $ aws iam list-access-keys --user-name compromised-user  
{  
  "AccessKeyMetadata": [  
    {  
      "UserName": "compromised-user",  
      "AccessKeyId": "AKIATPZ5Z47WIAQWLAVJ",  
      "Status": "Active",  
      "CreateDate": "2025-08-18T07:40:51+00:00"  
    }  
  ]  
}  
~ $ for AKI in $(aws iam list-access-keys --user-name compromised-user \  
> --query 'AccessKeyMetadata[].AccessKeyId' --output text); do  
> aws iam update-access-key --user-name compromised-user \  
> --access-key-id $AKI --status Inactive  
> done  
~ $ for AKI in $(aws iam list-access-keys --user-name compromised-user \  
> --query 'AccessKeyMetadata[].AccessKeyId' --output text); do  
> aws iam delete-access-key --user-name compromised-user --access-key-id $AKI  
> done  
~ $ aws iam delete-login-profile --user-name compromised-user
```

3) Restrict Permissions: Detach all policies from the user. Remove from any IAM group.

```
~ $  
~ $ for ARN in $(aws iam list-attached-user-policies --user-name compromised-user \  
> --query 'AttachedPolicies[].PolicyArn' --output text); do  
> aws iam detach-user-policy --user-name compromised-user --policy-arn $ARN  
> done  
~ $ for NAME in $(aws iam list-user-policies --user-name compromised-user \  
> --query 'PolicyNames[]' --output text); do  
> aws iam delete-user-policy --user-name compromised-user --policy-name $NAME  
> done  
~ $ for G in $(aws iam list-groups-for-user --user-name compromised-user \  
> --query 'Groups[].GroupName' --output text); do  
> aws iam remove-user-from-group --group-name $G --user-name compromised-user  
> done  
~ $
```

4) Enable MFA (Multi-Factor Authentication) on the account.



Task 3- Document the Incident Response Steps.

Incident Summary:

- **Incident Type:** Simulated account compromise.
- **Impact:** Unauthorized S3 deletion attempt blocked.
- **Detection Method:** AWS CloudTrail & CloudWatch alerts.
- **Response Actions:** Access keys deactivated, permissions removed, MFA enabled.